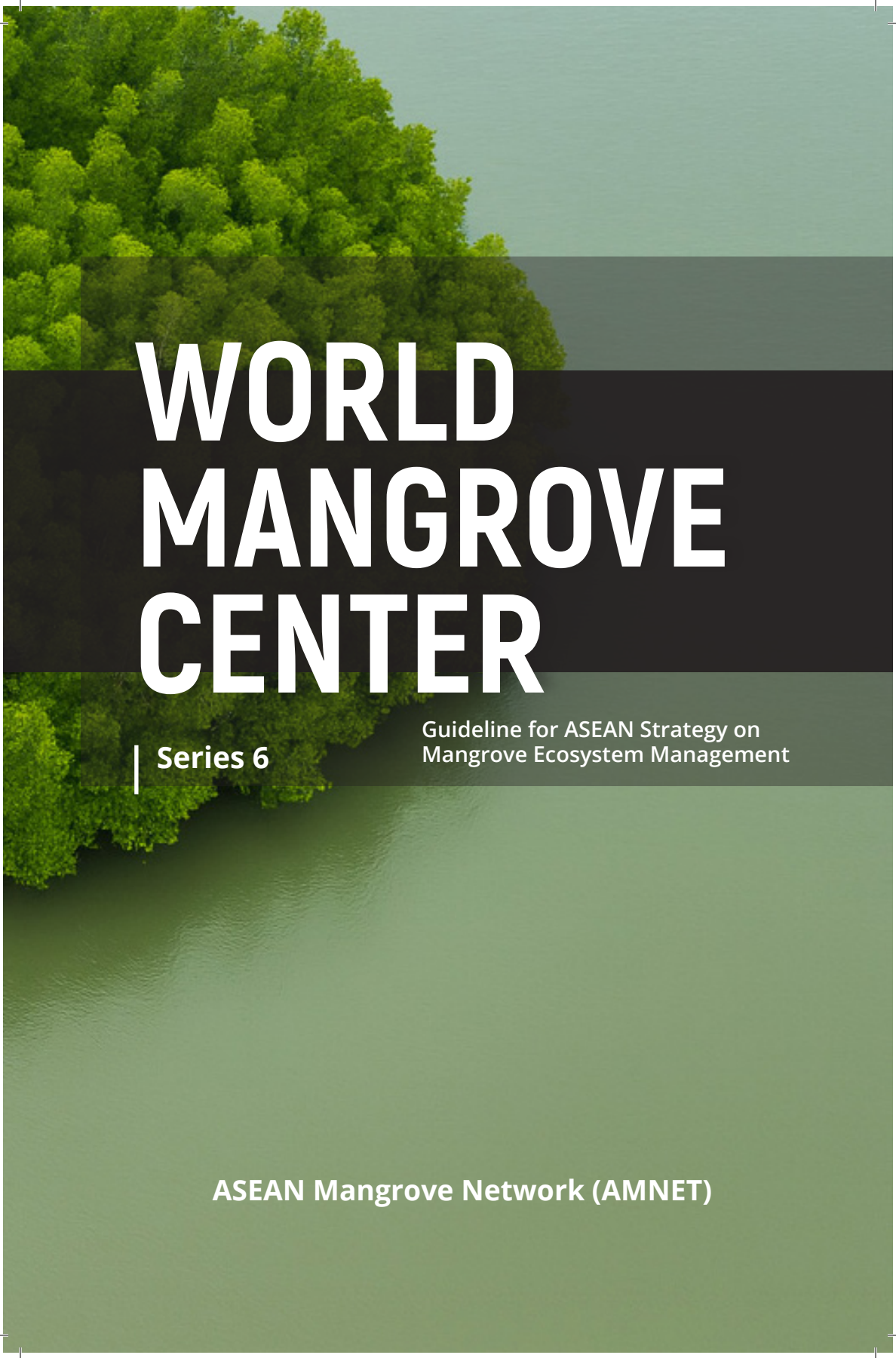

WORLD MANGROVE CENTER

Series 6

Guideline for ASEAN Strategy on Mangrove Ecosystem Management

ASEAN Mangrove Network (AMNET)

WORLD MANGROVE CENTER

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Series 6

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FOREWORD

The world is facing unprecedented environmental challenges—accelerating climate change, biodiversity loss, and ecosystem degradation. Among the most valuable yet vulnerable ecosystems are mangrove forests, often called the "rainforests of the sea." These coastal forests protect shorelines from erosion and storm surges, provide essential habitats for countless species, and serve as powerful carbon sinks that play a significant role in climate change mitigation.

Despite their immense ecological, economic, and cultural value, mangrove ecosystems continue to decline at an alarming rate due to land conversion, pollution, and unsustainable resource use, further compounded by the impacts of climate change. This reality underscores the urgent need for coordinated, science-based, and inclusive strategies to conserve and restore these vital ecosystems.

The ASEAN Mangrove Network (AMNET) serves as a key regional platform for advancing mangrove conservation and sustainable management. The development of this Technical Guideline for the World Mangrove Center reflects AMNET's collective commitment to safeguarding coastal ecosystems through regional cooperation, knowledge exchange, and shared stewardship.

This Guideline presents a comprehensive framework for the establishment and governance of the World Mangrove Center (WMC), built upon scientific evidence, traditional knowledge, and participatory approaches. It aligns with global frameworks such as the Sustainable Development Goals (SDGs) and the Paris

Agreement and supports ASEAN Strategy No. 4: *Implementing appropriate technical intervention and technology in conservation, protection, restoration, and sustainable use of mangrove ecosystems in each AMS.*

We extend our appreciation to the many experts, practitioners, policymakers, and community representatives whose contributions, insights, and dedication have shaped this document. Their collaborative efforts embody the spirit of unity and innovation needed to ensure lasting impact.

Author

Indonesia, June 9th 2025

I. INTRODUCTION

The ASEAN region boasts an extensive coastline of approximately 173,000 kilometers, making it one of the most coastal-oriented regions in the world. An estimated 77 percent of its total population, contributing to the broader 2.62 billion people living along the coastal belt in Asia-Pacific (PEMSEA, 2015), resides in areas directly adjacent to the sea. These coastal zones are not only densely populated but are also ecologically rich and economically vital.

Among the region's most important coastal ecosystems are mangrove forests, which span an estimated 4,512,000 hectares across ASEAN member states (Richards, Daniel R., Friess, Daniel A., *et al*., 2015). This area represents approximately 34 percent of the world's total mangrove coverage, positioning ASEAN as a global stronghold for mangrove biodiversity and functionality. These forests deliver critical ecological services such as carbon sequestration, shoreline stabilization, nursery grounds for fisheries, and disaster risk reduction, while also sustaining millions of livelihoods.

Despite their immense value, ASEAN's mangrove ecosystems have been steadily declining over the past two decades. The loss of mangrove cover is driven by a combination of factors that can be broadly categorized into natural phenomena (non-anthropogenic factors) and human-induced pressures (anthropogenic factors). Natural drivers include rising sea levels,

extreme weather events, and coastal erosion—all of which are intensifying due to climate change. On the other hand, anthropogenic threats such as land conversion for aquaculture, urban expansion, infrastructure development, pollution, and unsustainable harvesting have placed severe pressure on these ecosystems.

The continued degradation of mangroves not only threatens regional biodiversity but also undermines the climate resilience and food security of millions of people in ASEAN. This underscores the urgent need for integrated, science-based, and cross-sectoral approaches to mangrove conservation and restoration, alongside stronger policy coordination at the regional and national levels.

The sustainability of mangrove ecosystems is increasingly under threat due to a combination of anthropogenic disturbances (human-induced pressures) and natural factors (such as coastal erosion, storms, and sea-level rise). These threats have led to a significant and sustained decline in mangrove forest coverage globally and nationally.

According to data from the Food and Agriculture Organization (FAO, 2007), Indonesia experienced a 31% reduction in mangrove area between 1980 and 2005—a loss of approximately 1.3 million hectares, from an estimated 4.2 million hectares down to 2.9 million hectares. This alarming trend reflects both direct human activities such as land conversion for aquaculture, agriculture, and urban development, as well as the compounding effects of climate-related stressors.

On a global scale, the rate of mangrove loss has been gradually declining, but remains a serious concern. In the 1980s, the world lost an estimated 185,000 hectares of mangroves annually. This rate decreased to 118,500 hectares per year in the 1990s, and further declined to approximately 102,000 hectares annually

between 2000 and 2005. Despite this downward trend, the cumulative impact over several decades remains severe and largely irreversible in many locations.

More recent estimates from the Global Mangrove Alliance (GMA, 2021) indicate that mangrove areas have declined by approximately 4.3% over the past two decades. Although this percentage may appear modest, it represents the continued degradation of critical coastal ecosystems that play an essential role in climate mitigation, shoreline protection, and biodiversity conservation.

This persistent decline underscores the urgent need for comprehensive, science-based, and multi-stakeholder approaches to mangrove conservation and restoration. Without decisive action, the degradation of mangrove ecosystems will not only threaten ecological integrity but also diminish the livelihoods, food security, and climate resilience of millions of people, particularly in tropical coastal nations like Indonesia.

Table 1. Mangrove areas and loss in Southeast Asia 2000–2012

Country	Total mangrove in 2000, ha	Mangrove deforestation, ha	Mangrove habitat area lost, ha	Percentage of mangrove loss 2000–2012, %
Brunei Darussalam	11.054	48	41	0.37
Cambodia	47.563	1.218	1.086	2.28
Indonesia	2.788.683	60.906	48.025	1,72
Malaysia	557.805	18.836	15.809	2.83
Myanmar	502.466	27.957	27.770	5.53
Philippines	257.575	1.423	1.296	0.5
Singapore	583	–	–	–
Thailand	245.179	3.504	3.344	1.36
Viet Nam	215.154	531	528	0.25
Timor Leste	1.066	2	2	0.19
Total Southeast Asia	4.626.545	114.424	97.901	2.12

Source: Rates and drivers of mangrove deforestation in Southeast Asia, 2000-2012 (Richards, Daniel R., Friess, Daniel A., et al. 2015)

Between 2000 and 2020, the primary direct causes of global mangrove loss were predominantly driven by human activities and natural processes. Aquaculture development was the leading factor, responsible for approximately 27 percent of the total mangrove area lost, closely followed by natural retraction, which accounted for 26 percent. Other significant contributors included the conversion of mangrove forests to oil palm plantations and rice cultivation, each representing about 8 percent of the loss, as well as other agricultural activities which made up around 12 percent. Notably, the impact of aquaculture on mangrove deforestation decreased over the two decades, dropping from roughly 31 percent in the first decade (2000–2010) to 21 percent in the second (2010–2020).

In contrast, the conversion of mangroves to oil palm plantations saw a dramatic increase during the same period, rising from about 4 percent in the early 2000s to 14 percent by the 2010s, with most of this activity concentrated in Southeast Asia. On a more positive note, natural expansion was responsible for the majority of mangrove gains worldwide, contributing to 82 percent of the total area recovered between 2000 and 2020, while active restoration efforts accounted for the remaining portion. This dynamic highlights the complex balance of losses and gains shaping the state of mangrove ecosystems globally, driven by both anthropogenic pressures and natural regeneration processes.

World Mangrove Center is critically important as it serves as a global hub for the conservation, research, and sustainable management of mangrove ecosystems, which are among the most productive and biologically diverse coastal habitats on Earth. Mangroves provide essential services such as protecting shorelines from erosion and storm surges, acting as natural

buffers against the impacts of climate change, and supporting rich biodiversity, including many endangered species. Additionally, mangroves play a significant role in carbon sequestration, capturing and storing large amounts of carbon dioxide, thereby helping to mitigate global warming. The center's efforts in education, community engagement, and policy advocacy promote awareness and encourage sustainable practices that balance environmental preservation with the livelihoods of local communities who depend on these ecosystems. By fostering scientific research and international collaboration, the World Mangrove Center not only helps to restore degraded mangrove forests but also strengthens resilience against environmental threats, ensuring that these vital ecosystems continue to provide critical ecological, economic, and social benefits worldwide.

World Mangrove Center (WMC) is a strategic national initiative aimed at developing a center for conservation, research, education, and international collaboration related to mangrove ecosystems. Through these multifaceted roles, the World Mangrove Center significantly contributes to the preservation and sustainable management of mangrove ecosystems, benefiting both the environment and society.

The success of WMC heavily relies on the implementation of advanced technology standards and the provision of adequate supporting facilities. The importance of WMC can be seen from several aspects:

1. Ecological Aspect**:** WMC plays a vital role in maintaining coastal resilience by protecting shorelines from erosion and extreme weather events. It also contributes to blue carbon storage, which is crucial for mitigating climate change, and serves as a habitat for diverse biodiversity, supporting numerous marine and terrestrial species.

2. Economic Aspect**:** WMC helps generate income through eco-tourism activities that attract visitors interested in natural and sustainable experiences. Additionally, it supports carbon markets by facilitating the trading of carbon credits derived from mangrove conservation and restoration efforts.
3. Social Aspect**:** The center empowers coastal communities by providing education, training, and resources that enhance local capacity and improve livelihoods, promoting sustainable use and stewardship of mangrove ecosystems.
4. Policy Aspect**:** WMC supports Indonesia's commitments to international frameworks such as the Nationally Determined Contributions (NDCs) under the Paris Agreement, the Sustainable Development Goals (SDGs), and environmental diplomacy, strengthening the country's leadership in global environmental governance.
5. Scientific Aspect: As a national data center, WMC acts as a hub for collecting, managing, and sharing mangrove-related information. It also fosters international research collaboration, facilitating scientific advancements that inform policy and conservation strategies.

II.

CONCEPT AND ROLE OF THE WORLD MANGROVE CENTER

A. Definition and Main Functions

World Mangrove Center (WMC) is a dedicated institution focused on the conservation, research, education, and sustainable management of mangrove ecosystems on a global scale. Mangroves are unique coastal forests found in tropical and subtropical regions that serve as vital ecological buffers between land and sea. The WMC embodies a comprehensive approach to addressing the environmental, economic, social, and scientific importance of mangroves. It acts as a central hub for coordinating efforts to protect these fragile ecosystems from threats such as deforestation, climate change, and unsustainable development. The center promotes the integration of advanced technologies and scientific research to better understand mangrove dynamics, carbon sequestration potential, biodiversity conservation, and the role of mangroves in supporting coastal communities. Beyond environmental protection, the WMC fosters collaboration among governments, researchers, local communities, and international organizations to implement effective policies and practices that balance ecological preservation with economic development, such as eco-tourism and carbon trading. Additionally, the center serves as an educational platform to raise global awareness about the critical functions of mangroves in climate mitigation,

disaster risk reduction, and livelihood support. By uniting diverse stakeholders and providing evidence-based solutions, the World Mangrove Center plays an essential role in ensuring the resilience and sustainability of mangrove ecosystems for current and future generations.

Indonesia is the world's biggest owner of mangrove cover (3.36 million ha). About 45% are in Papua, 22% in Kalimantan, 20% in Sumatra, 6% in Maluku, and 4% in Sulawesi, while the rest are spread over Java, Bali, and Nusa Tenggara. Indonesia has a high diversity of flora in its mangrove ecosystem. Indonesia also has a fairly high diversity of fauna. 50 mangrove species exist throughout the world, 43 of them can be found in Indonesia.

At the international level, Indonesia played a leading role by proposing the UNEA Resolution on the Sustainable Management for the Global Health of Mangroves, which was successfully adopted during the fourth session of the United Nations Environment Assembly (UNEA 4) held in Nairobi in 2019. Building on this achievement, the Indonesian government took the initiative to establish the World Mangrove Center in 2021. This center serves as a center of excellence and an information hub dedicated to advancing knowledge and best practices in the management of mangrove ecosystems. The creation of the World Mangrove Center represents a comprehensive effort to address the complex challenges involved in mangrove management, including the restoration of degraded mangrove areas and the promotion of a global movement focused on conservation, rehabilitation, and sustainable use of mangroves.

Indonesia's commitment to founding this center not only strengthens regional approaches toward managing mangrove ecosystems but also provides a platform to elevate its impact

more broadly within the ASEAN region, encouraging cooperation and collaboration among member countries. This initiative underscores Indonesia's leadership in fostering sustainable environmental practices and highlights the importance of mangroves for ecological resilience and community well-being on a regional and global scale.

B. WMC Is an International Hub

According to the Ministry of Environment and Forestry of the Republic of Indonesia in its presentation in Kyrgyzstan in 2024, the World Mangrove Center can be an international Hub that can divide into 3 purpose.

1. The first, WMC aims to provide impactful, transformative, and action-oriented evidence along with practical solutions specifically designed to address and reduce the main drivers of mangrove deforestation and degradation. In doing so, the Center strives to ensure the long-term preservation and sustainable management of these vital ecosystems, which play a crucial role in maintaining coastal resilience, supporting biodiversity, and mitigating the impacts of climate change. Furthermore, the Center seeks to empower local communities by strengthening their capacity to sustainably manage mangrove resources, promoting economic opportunities such as ecotourism and carbon trading, and fostering international collaboration to exchange knowledge and best practices. Ultimately, the goal of the World Mangrove Center is to become a global leader and international hub for mangrove conservation efforts, generating positive environmental and social impacts for both current and future generations.

2. The Second is the vision. World Mangrove Center (WMC) envisions becoming a leading international hub for knowledge exchange, scientific research, policy dialogue, and multilateral cooperation dedicated to the sustainable management and protection of mangrove ecosystems. Its vision is rooted in the urgent need to halt the widespread conversion and degradation of mangroves, which continue to be threatened by unsustainable development, aquaculture, agriculture, and climate change. By serving as a center of excellence, WMC aspires to facilitate the sharing of best practices, innovative technologies, and evidence-based strategies that support effective conservation, restoration, and sustainable utilization of mangrove resources. It also aims to build strong partnerships across countries, institutions, and communities, fostering global solidarity and joint actions to safeguard mangroves. Ultimately, WMC's vision reflects a commitment to preserving the ecological, social, and economic value of mangroves, ensuring their resilience and benefits for both present and future generations.
3. The Third is the Mission. World Mangrove Center (WMC) carries a mission to bring together a diverse range of stakeholders—including governments, researchers, local communities, non-governmental organizations, private sector actors, and international institutions—to collaboratively advance scientific understanding, co-develop innovative solutions, and facilitate the exchange of knowledge for practical implementation.

This mission is driven by the urgent need to protect and restore mangrove ecosystems, which are not only rich in biodiversity but also vital in supporting the livelihoods of millions of people and serving as natural defenses against the impacts of climate

change. WMC aims to bridge the gap between research and practice by translating data, evidence, and traditional knowledge into actionable policies and programs that enhance the resilience of mangrove forests and the well-being of mangrove-dependent communities. Through capacity-building, inclusive dialogue, and international cooperation, WMC seeks to create a unified global movement that promotes the sustainable management of mangroves as a key nature-based solution in the face of the growing climate crisis.

C. Strategic Benefits for ASEAN Country

The establishment of the World Mangrove Center (WMC) offers significant strategic benefits for ASEAN countries, both at regional and international levels. As many ASEAN member states possess extensive mangrove ecosystems and rely heavily on coastal resources for economic and social development, WMC serves as a vital platform to enhance collective efforts in managing these critical environments. Below are some key strategic benefits for ASEAN:

1. Regional Collaboration and Integration

   WMC provides a centralized platform for ASEAN countries to collaborate in mangrove conservation, research, and restoration. By facilitating dialogue, knowledge-sharing, and joint action, the Center helps strengthen regional unity and fosters a coordinated response to shared environmental challenges, such as climate change and coastal degradation.

2. Capacity Building and Technology Transfer

   Through training programs, workshops, and technical assistance, WMC contributes to enhancing the technical capacities of ASEAN member states. It enables the exchange

of expertise and the transfer of best practices and innovative technologies, helping countries improve their mangrove monitoring, restoration, and management systems.

3. Supporting Regional Environmental Goals

   WMC directly supports ASEAN's environmental priorities, including commitments under the ASEAN Working Group on Coastal and Marine Environment (AWGCME), ASEAN Agreement on the Conservation of Nature and Natural Resources, and regional action plans for biodiversity and climate adaptation. It also aligns with global frameworks like the SDGs and the Paris Agreement.

4. Promoting Sustainable Blue Economy

   The Center plays a role in advancing the ASEAN agenda on the sustainable blue economy by encouraging mangrove-based livelihoods, eco-tourism, and carbon credit markets. This contributes to inclusive economic development while maintaining the ecological integrity of coastal zones.

5. Disaster Risk Reduction and Climate Resilience

   By supporting mangrove conservation and restoration, WMC strengthens the natural resilience of coastal areas, reducing vulnerability to tsunamis, storm surges, sea-level rise, and erosion—all of which are growing threats for ASEAN countries due to climate change.

6. Elevating ASEAN's Global Leadership

   WMC positions ASEAN as a proactive regional bloc in addressing global environmental issues. It provides a platform for ASEAN countries to showcase their leadership in nature-based solutions, environmental diplomacy, and south-south cooperation, thereby enhancing their influence in global climate and biodiversity negotiations.

In summary, the World Mangrove Center not only benefits individual ASEAN countries through improved ecosystem management and community empowerment but also enhances the region's strategic position in shaping sustainable environmental governance at the global level.

On the other hand, there are additional benefits that ASEAN countries can gain through the development of the World Mangrove Center (WMC). One of the key advantages is

1. Opportunity to foster stronger regional cooperation in mangrove management. Through WMC, ASEAN member states can collaborate more effectively to develop and implement joint strategies that address shared challenges, while maximizing the potential of mangroves as valuable natural assets. This cooperation can generate a wide range of social, economic, and environmental benefits, including the empowerment of coastal communities, the promotion of sustainable livelihoods such as eco-tourism and mangrove-based fisheries, and the enhancement of ecosystem services like carbon sequestration, water filtration, and shoreline protection. Furthermore, shared access to data, research findings, and restoration technologies through the WMC can elevate regional capacity, reduce duplication of efforts, and accelerate progress toward national and ASEAN-wide environmental goals. In essence, the WMC becomes not just a conservation platform, but a catalyst for inclusive growth and resilience across the region.
2. Based on the Government of Indonesia through the Ministry of Environment and Forestry in 2024, mangrove ecosystem management in Indonesia holds significant potential to serve as a platform for the exchange of experts, knowledge, and practical experiences ranging from success stories to lessons learned that can be adapted and applied across

various regional and global contexts. Given Indonesia's extensive experience in mangrove restoration, community-based management, and integrated coastal governance, the country can play a leading role in facilitating knowledge-sharing initiatives that benefit other nations facing similar challenges.

This exchange may include collaborative training programs, expert missions, workshops, and field visits that showcase effective strategies and locally driven solutions. Furthermore, Indonesia's applied research outcomes—such as innovations in mangrove monitoring, restoration techniques, blue carbon accounting, and ecosystem-based adaptation—can provide valuable insights and scalable models for use in diverse ecological and socio-political settings. Through such exchanges, Indonesia contributes to strengthening global and regional capacities in sustainable mangrove management while promoting mutual learning, cooperation, and the harmonization of conservation efforts in line with broader environmental and climate goals;

3. Data and information related to the sustainable management of mangroves, as well as the outcomes of associated economic development initiatives, which are collected, curated, and analyzed by the World Mangrove Center (WMC), have the potential to serve as valuable references at both regional and global levels. This comprehensive repository of knowledge can inform evidence-based policymaking, guide restoration practices, and support the design of sustainable livelihood programs in other countries facing similar environmental and socio-economic challenges. By providing access to reliable data on ecosystem services, biodiversity, carbon stock assessments, socio-economic

impacts, and best practices in mangrove governance, WMC can act as a knowledge hub that empowers decision-makers, researchers, and practitioners worldwide. In addition, this centralized information source enhances transparency, supports monitoring and evaluation efforts, and fosters international cooperation by aligning strategies with global frameworks such as the Sustainable Development Goals (SDGs), the Paris Agreement, and the Post-2020 Global Biodiversity Framework, and

4. A country can transform its achievements in sustainable mangrove management into a powerful showcase for the world, demonstrating effective conservation practices and sustainable development models that others can learn from and emulate. By highlighting successful restoration projects, community-based management initiatives, and innovative policy frameworks, the country not only elevates its environmental leadership but also inspires global efforts to protect and sustainably use mangrove ecosystems. This "show window" serves as a platform to share best practices, promote eco-friendly economic opportunities such as carbon trading and eco-tourism, and attract international partnerships and investments.

   Ultimately, by turning its sustainable mangrove management success into a global example, the country contributes significantly to raising awareness about the critical importance of mangroves for climate resilience, biodiversity conservation, and the livelihoods of millions worldwide.

D. The Importance of the World Mangrove Center for ASEAN Policy Frameworks

The World Mangrove Center (WMC) holds significant importance for both national and ASEAN policy frameworks by serving as a central institution that supports and strengthens the development and implementation of mangrove-related policies across multiple levels. At the national level, the WMC provides critical scientific data, technical expertise, and practical guidance that inform policy decisions aimed at conserving, restoring, and sustainably managing mangrove ecosystems. These ecosystems are vital for coastal protection, carbon sequestration, biodiversity preservation, and supporting the livelihoods of millions of people, particularly in countries like Indonesia with extensive mangrove coverage. By acting as a hub for research, innovation, and knowledge exchange, the WMC helps policymakers craft effective strategies that align with national commitments to climate change mitigation, biodiversity conservation, and sustainable development goals.

On a regional scale, the WMC plays a pivotal role in harmonizing mangrove policies among ASEAN member states, fostering greater cooperation and coordination to address shared environmental challenges. Mangrove ecosystems often span national boundaries, and the threats they face—such as deforestation, pollution, and climate change—require collective action. The Center facilitates dialogue and collaboration, enabling ASEAN countries to share best practices, develop unified approaches, and implement regional programs that enhance the resilience of coastal communities and ecosystems. Furthermore, the WMC supports ASEAN's environmental and

climate objectives by providing evidence-based inputs that strengthen regional policy frameworks and promote nature-based solutions as integral components of climate adaptation and mitigation strategies

In addition, the World Mangrove Center serves as a platform for ASEAN to amplify its voice and leadership in global environmental governance. By showcasing successful regional initiatives and demonstrating commitment to sustainable mangrove management, ASEAN can influence international discussions and partnerships related to climate change, biodiversity, and sustainable development. Ultimately, the WMC not only enhances national and regional policy coherence but also contributes to building a resilient and sustainable future for mangrove ecosystems and the communities that depend on them across Southeast Asia.

Generally, the importance of the World Mangrove Center (WMC) for ASEAN country policies can divide into four key points:

1. Becoming a National Center of Excellence for Science-Based Mangrove Management

1. Indonesia hosts the largest mangrove forest coverage in the world; over 3.4 million hectares, with exceptionally high biodiversity. However, challenges persist, including ecosystem degradation, land-use conversion, and data and capacity gaps across regions.

   The World Mangrove Center will serve as:

   - A national center of excellence for data-driven and science-based mangrove policy development
   - A facility that integrates spatial, ecological, social, and economic data to support policy formulation

- » A reference for restoration standards and protocols, accessible to local governments, research institutions, and community groups
- » hub for training, joint research, and local innovation development (e.g., mangrove-based bioeconomy)

2. Strengthening Indonesia's Environmental Diplomacy and Leadership in ASEAN

   The ASEAN region holds more than 40% of the world's mangroves, yet faces intense pressure from coastal development, reclamation, and climate impacts. To date, there is no dedicated regional coordination center for mangrove ecosystems.

   With the WMC based in Indonesia, it can serve as:

   - » A regional hub for research, training, and technical cooperation in mangrove management
   - » A platform for developing and harmonizing regional standards and guidelines for restoration and conservation
   - » A center for transboundary cooperation on shared mangrove ecosystems and major river deltas
   - » A symbol of Indonesia's leadership in advancing climate and environmental goals within ASEAN

   Relevance to ASEAN policies:

   The WMC aligns with the ASEAN Blue Economy Framework**,** ASEAN Strategic Plan on Environment, and supports ASEAN's contributions to the Sustainable Development Goals (SDGs)**–** especially Goal 13 (Climate Action) and Goal 14 (Life Below Water)**.**

3. Bridging Data and Capacity Gaps at Regional and Global Levels

   Many ASEAN countries face challenges in terms of data availability, technical capacity, and ecosystem monitoring systems. The WMC can play a critical role by:

   - Establishing a centralized mangrove information and data repository, including blue carbon data, land cover change, and climate vulnerability
   - Developing and sharing standardized methodologies for mangrove mapping, inventories, and monitoring
   - Serving as a platform for technical training and capacity-building, including South-South knowledge exchanges
   - Acting as a bridge among researchers, policymakers, and civil society in collaborative coastal governance

4. Responding to the Climate Crisis and Advancing Nature-Based Solutions

   Mangroves are among the most climate-resilient ecosystems, offering both high carbon storage potential and natural protection from climate-related disasters (e.g., erosion, storm surges). Yet their value is still underrepresented in policy and finance systems.

   The World Mangrove Center will contribute by:

   - Mainstreaming mangroves as a key nature-based solution (NbS) for climate adaptation and mitigation
   - Promoting payment for ecosystem services (PES), blue carbon finance, and green economic incentives
   - Providing scientific and technical evidence to unlock public and private investment in mangrove sectors
   - Supporting an inclusive blue economy framework across ASEAN that emphasizes ecological and community resilience

2. Scope of World Mangrove Center

1. Encouraging international sharing of data and information, knowledge, and best practices in mangrove management

   World Mangrove Center (WMC) will serve as an open platform for countries, research institutions, and the global community to share spatial data, scientific research results, and field experiences in mangrove management. Its functions include:

   - Providing open access to a global mangrove database (covering area, ecosystem conditions, blue carbon, etc.);
   - Facilitating cross-country learning on restoration techniques, conservation, and monitoring;
   - Supporting data integration into evidence-based policymaking

2. Developing knowledge sharing and capacity building

   WMC will organize various programs to strengthen human and institutional capacity, including:

   - Technical training (e.g., restoration techniques, carbon monitoring, nature-based solutions)
   - Workshops, seminars, and international conferences
   - Expert exchange programs and regional/international internships
   - Empowering local communities to actively participate in community-based mangrove management

3. Developing international reference of restoration, rehabilitation, and conservation with consideration of ecological, social, and economic aspects, especially in facing the climate crisis (adaptation and mitigation)

WMC will be a comprehensive and adaptive global model that:

- » Integrates a multi-dimensional approach: ecological sustainability, social inclusion, and economic wellbeing of coastal communities
- » Provides guidelines and models for science-based and socially just mangrove restoration
- » Positions mangroves as nature-based solutions (NbS**)** for climate change mitigation (carbon storage) and adaptation (coastal protection, alternative livelihoods)

4. Encourage mangrove management standards based on science and best practices

   WMC will develop and disseminate international standards that can be used for:

   - » Evaluating the success of mangrove restoration projects
   - » Scientific monitoring and evaluation protocols based on measurable indicators
   - » Guidelines for sustainable mangrove management adaptable to local contexts
   - » Alignment with global standards such as Ramsar, IPCC (blue carbon), and FAO frameworks

5. Encouraging international dialogue and cooperation for mangrove management

   WMC will act as an environmental diplomacy platform and a global network facilitator by:

   - » Hosting an annual international forum involving policymakers, researchers, and civil society
   - » Initiating bilateral/multilateral cooperation among coastal nations (e.g., South-South Cooperation, Mangrove Alliance for Climate)

- Establishing partnerships with NGOs, the private sector, international financial institutions, and UN agencies
- Strengthening regional collaboration in Southeast Asia, the Pacific, and major delta nations

6. Promote sustainable mangrove management

   WMC aims to promote long-term sustainable mangrove management systems that:

   - Balance conservation with responsible use (e.g., ecotourism, fisheries, bioenergy)
   - Enhance coastal community livelihoods through mangrove-based solutions
   - Support green economic incentives, such as payment for ecosystem services and blue carbon certification
   - Increase the resilience of ecosystems and communities against climate change and natural disasters

Figure 1. Scope of WMC

III.

PLANNING AND DEVELOPMENT OF THE WORLD MANGROVE CENTER (WMC)

The planning and development of the World Mangrove Center (WMC) represent a comprehensive and strategic effort to establish a leading institution dedicated to the conservation, restoration, and sustainable management of mangrove ecosystems globally. This process involves a multi-phase approach that begins with identifying the critical needs and challenges faced by mangrove habitats, particularly in regions such as Southeast Asia where these ecosystems are both rich in biodiversity and under significant threat from human activities and climate change.

In the initial planning phase, extensive consultations are conducted with a broad range of stakeholders, including government agencies, scientific institutions, local communities, non-governmental organizations, and international partners. These consultations help define the core objectives, scope, and functions of the Center, ensuring that it addresses both local and global priorities. Key considerations include the integration of cutting-edge scientific research, the incorporation of traditional ecological knowledge, and the alignment with international environmental frameworks such as the Sustainable Development Goals (SDGs), the Paris Agreement, and the Post-2020 Global Biodiversity Framework.

The development phase focuses on building the institutional capacity and infrastructure necessary for the WMC to function effectively. This includes establishing physical facilities, such as research laboratories, training centers, and data repositories, as well as creating robust governance structures that promote transparency, accountability, and inclusive decision-making. Emphasis is placed on fostering strong partnerships across countries and sectors to leverage expertise, share resources, and coordinate efforts on mangrove conservation and sustainable use.

Furthermore, the WMC's development incorporates advanced technological tools for monitoring and managing mangrove ecosystems, such as remote sensing, geographic information systems (GIS), and bioinformatics. These technologies enhance the Center's ability to generate high-quality data, track changes over time, and assess the effectiveness of restoration and protection initiatives.

Capacity building and community engagement are also central to the planning and development of the WMC. The Center aims to empower local populations by providing education, training, and economic opportunities linked to mangrove conservation, thereby ensuring that sustainable management practices are both socially inclusive and economically viable.

Overall, the planning and development of the World Mangrove Center is designed to create a world-class hub that not only advances scientific knowledge and policy innovation but also drives on-the-ground action to protect mangroves. By doing so, the WMC contributes to global efforts to mitigate climate change, preserve biodiversity, and support the resilience and well-being of coastal communities.

A. Initiation & Preliminary Assessment (Preliminary Stage)

This initial phase sets the foundation by thoroughly understanding the environmental, social, and legal landscape where the WMC will be established. Identifying a strategic location ensures the center's relevance and accessibility. Stakeholder mapping helps build inclusive cooperation for sustainable management.

Output: Feasibility Study & Academic Manuscript

- Identify strategic locations based on ecological value, social aspects, and accessibility.
- Conduct assessments of potential and threats, including biodiversity status, ecosystem degradation, and surrounding communities.
- Review regulatory and legal frameworks concerning the designated area.
- Perform stakeholder analysis involving government agencies, indigenous communities, NGOs, academia, and private sectors.

B. Development of Grand Design/Masterplan

The masterplan details the long-term vision and operational blueprint of the WMC, ensuring the center's activities are purpose-driven and aligned with broader development and climate commitments. Proper zoning balances ecological protection with public use and scientific activities.

Output: WMC Masterplan Document

- Define vision, mission, goals, and branding for the Center.
- Zone the area for different functions such as research, conservation, education, tourism, and buffer zones.

- Design green infrastructure that is environmentally friendly and climate resilient.
- Project potential economic, social, and environmental impacts.
- Integrate with national and international policies and frameworks such as SDGs, Nationally Determined Contributions (NDC), Strategic Environmental Assessments (KLHS), National Medium-Term Development Plans (RPJMN), and the Mangrove for Climate Alliance.

C. Establishment of Managing Institution (Governance Model)

Creating a robust governance framework ensures the WMC operates efficiently and inclusively. Diversified funding sources and clear accountability mechanisms support sustainable and transparent management.

The establishment of a managing institution, or governance model, is essential for the World Mangrove Center (WMC) because it provides a clear and organized framework for leadership, decision-making, and coordination of activities. A well-defined governance structure ensures that roles and responsibilities among government bodies, academic institutions, local communities, and private sectors are clearly delineated, promoting effective collaboration and reducing conflicts. This organizational clarity is critical for managing complex tasks such as funding allocation, program implementation, and compliance with legal and environmental regulations. Moreover, a strong governance model supports transparency and accountability by establishing monitoring and reporting mechanisms, which are vital for building trust among stakeholders and securing continued financial and political support. It also enables the WMC to respond adaptively to emerging challenges, coordinate

multi-sectoral efforts, and maintain long-term sustainability. Ultimately, the managing institution acts as the backbone of the WMC, ensuring that its mission to conserve and sustainably manage mangrove ecosystems is carried out efficiently, inclusively, and with measurable impact.

Output: Management Body or Special Unit

- Define organizational structure clarifying the roles of government, academia, private sector, and local communities.
- Develop funding mechanisms through government budgets (APBN), Corporate Social Responsibility (CSR), green investments, and international cooperation.
- Establish monitoring, reporting, and accountability systems to ensure transparency and effectiveness.

D. Physical Development and Supporting Systems

Building state-of-the-art facilities and digital infrastructure equips the WMC with the necessary tools for cutting-edge research, education, and real-time ecosystem monitoring, enhancing its capacity to manage mangrove ecosystems effectively.

Output: Physical & Digital Facilities

- Construct research centers, education rooms, mangrove tracking stations, docks, and observatories.
- Develop data centers, information platforms, and ecosystem monitoring systems.
- Integrate advanced technologies such as Artificial Intelligence (AI) for mangrove monitoring, drone mapping, and spatial information systems (GIS).

E. Operationalization & Program Activation

This phase activates the core functions of the WMC, translating planning into practical programs that support ecosystem health and community engagement while promoting sustainable economic activities. Operationalization and program activation are vital for the World Mangrove Center (WMC) because they transform plans and strategies into tangible actions that directly benefit ASEAN countries. This phase ensures that research, conservation, education, and community engagement initiatives move beyond theory and become practical efforts that protect and restore mangrove ecosystems. For ASEAN nations, many of which rely heavily on coastal resources, activating programs such as mangrove rehabilitation, eco-tourism development, and capacity-building helps safeguard biodiversity, enhance coastal resilience against climate change, and improve local livelihoods. Operationalization also fosters regional collaboration by enabling the sharing of successful approaches and technologies tailored to ASEAN's diverse ecological and socio-economic contexts. Furthermore, active programs generate data and lessons learned that inform policy-making and adaptive management across member countries. In essence, without operationalization and program activation, the World Mangrove Center's vision would remain abstract; this stage is what drives real environmental, social, and economic impacts, reinforcing ASEAN's commitment to sustainable development and regional cooperation in mangrove conservation.

Output: Ongoing Activities

- Conduct multi-scale mangrove research at local, national, and global levels.
- Implement community and school education programs.

- Develop conservation-based tourism (ecotourism) initiatives.
- Launch conservation and rehabilitation programs including planting, nursery management, and bioengineering techniques.

F. Monitoring, Evaluation, and Replication

Regular monitoring and evaluation enable adaptive management and demonstrate the Center's impact. Sharing successful approaches encourages scaling and replication, multiplying benefits across Indonesia and beyond. Monitoring, evaluation (include reporting) are crucial components in the management of the World Mangrove Center (WMC) because they ensure that the center's activities are effective, transparent, and aligned with its goals. Monitoring provides real-time data on the progress of various programs, such as conservation efforts, restoration projects, and community engagement initiatives. This ongoing observation helps identify any challenges or deviations early, allowing for timely corrective actions. Evaluation, on the other hand, assesses the overall impact and outcomes of these activities against predefined objectives and sustainability indicators. It helps determine whether the strategies implemented are achieving desired environmental, social, and economic results.

Reporting is equally important because it communicates these findings to stakeholders, including government agencies, local communities, funding partners, and international collaborators. Transparent reporting builds trust, accountability, and support for the WMC's mission. Furthermore, regular monitoring, evaluation, and reporting create a feedback loop that informs adaptive management, enabling the center to refine its approaches, scale successful initiatives, and replicate best practices in other regions. Ultimately, this systematic process

ensures the long-term sustainability and credibility of the WMC, enhancing its role as a global leader in mangrove conservation and management.

Output: Impact Reports & Replication Models

- Conduct evaluations based on SDG indicators and sustainability criteria.
- Report outcomes to government bodies and international partners.
- Develop replication schemes to apply successful models in other regions such as Papua, Sumatra, and Sulawesi.

Table 2. Summary Framework of WMC Roadmap

Phase	Key Outputs	Core Activities	Purpose
1. Initiation & Preliminary	Feasibility study & academic paper	Location identification, stakeholder analysis, regulation review	Lay foundational understanding & stakeholder engagement
2. Grand Design/ Masterplan	Masterplan document	Vision/mission setting, zoning, infrastructure design, policy integration	Strategic direction and policy alignment
3. Governance Model	Management institution established	Organizational setup, funding mechanism, accountability system	Sustainable, transparent institutional management
4. Physical & Digital Development	Research & monitoring facilities	Build facilities, implement tech systems	Equip WMC with tools for research, education, and monitoring
5. Operationalization	Active programs	Research, education, ecotourism, conservation	Deliver practical impact and community engagement
6. Monitoring & Replication	Impact reports, replication plans	Evaluation, reporting, scale-up models	Ensure sustainability, accountability, and broader impact

IV.

INTEGRATED INFRASTRUCTURE AND INFORMATION SYSTEM OF THE WORLD MANGROVE CENTER

The World Mangrove Center (WMC) plays a critical role in the global effort to conserve and sustainably manage mangrove ecosystems. These ecosystems provide a wide range of environmental, economic, and social benefits—from acting as natural coastal barriers and carbon sinks to supporting biodiversity and the livelihoods of local communities. Given the scale and complexity of these tasks, having a strong and reliable infrastructure and support system, particularly a robust data and information system, is not just beneficial but essential.

A. Centralization of Global Mangrove Data

One of the greatest challenges in mangrove conservation is the fragmentation of data. Countries and organizations collect different kinds of information using varied methodologies, resulting in inconsistencies. A centralized system at WMC serves as a single point of reference for all stakeholders. It consolidates data from multiple sources—national governments, local communities, research institutions, and international organizations—into a unified platform. This allows for better coordination, minimizes duplication of effort, and enhances transparency.

B. Standardization for Consistency and Quality

The WMC's data infrastructure enables the standardization of information, ensuring that data is collected, stored, and analyzed using consistent criteria. This is vital for:

- Comparing mangrove health and coverage across regions.
- Tracking progress of conservation efforts.
- Ensuring reliability and scientific validity of information.

Standardized data supports evidence-based policymaking, improves international collaboration, and facilitates compliance with global environmental agreements.

C. Efficient Data Input and Updating Mechanism

The system allows designated administrators from both countries and the WMC Secretariat to input and update data. This ensures that information is kept current and reflective of real-time developments on the ground. Such flexibility allows the system to handle both static data (e.g., institutional profiles) and dynamic data (e.g., mangrove coverage maps, publications, non-timber forest product inventories).

D. Support for Decision-Making and Policy Development

- An effective data and information system provides stakeholders—including governments, NGOs, researchers, and funding bodies—with accurate and timely data. This enables:

- Informed policy development at national and international levels.
- Targeted interventions for areas under threat.
- Justification for funding and resource allocation.
- Design of education and awareness programs based on scientific facts.

E. Research and Innovation

A well-maintained database opens doors for advanced research. Scholars and scientists can access a wealth of information needed to explore:

- Climate change impacts on mangrove ecosystems.
- Species diversity and ecological interactions.
- Socio-economic dimensions of mangrove use and conservation.

It also fosters innovation in restoration techniques, community-based management models, and adaptive policy frameworks.

F. Transparency and Accessibility via Dashboard System

The integration of a dashboard system, especially one based at a central location like Bali, ensures that processed data is not only secure but also accessible and user-friendly. Visualizations, maps, and reports can be generated to make complex data understandable for decision-makers and the public alike. Transparency builds trust and encourages greater stakeholder engagement.

G. Strengthening International Collaboration

The WMC acts as a global hub with a reliable information system, it becomes easier to coordinate multinational projects, facilitate knowledge exchange, and promote best practices. Countries can benchmark their efforts, identify gaps, and align strategies to achieve shared global targets such as the UN Sustainable Development Goals (SDGs) and the UN Decade on Ecosystem Restoration.

H. Risk Management and Early Warning

Risk management and early warning systems are critical components of the World Mangrove Center's (WMC) mandate to protect and sustainably manage global mangrove ecosystems. As mangroves face increasing threats from climate change, coastal development, pollution, and illegal activities, WMC must be equipped with the tools and strategies to anticipate, respond to, and mitigate these risks effectively.

An effective data infrastructure also plays a crucial role in disaster preparedness and risk management. Real-time monitoring of mangrove ecosystems can provide early warnings about degradation, pollution, or threats such as illegal logging and coastal development.

1. Purpose and Importance

The primary goal of risk management and early warning systems at WMC is to:

- Prevent or minimize damage to mangrove ecosystems.
- Provide timely alerts to stakeholders (governments, local communities, conservation agencies) about potential or ongoing threats.

- Improve responsiveness and resource allocation during environmental crises.
- Support proactive planning and long-term resilience strategies.

2. Core Components of the System

a) Real-Time Monitoring

» WMC's information infrastructure is expected to integrate real-time monitoring technologies, such as:

» Satellite imagery for detecting deforestation, fires, and land-use change.

» Remote sensors for tracking water quality, salinity, and tidal patterns.

» Drones and ground-based observations for rapid assessment.

b) Data Analytics and Predictive Modelling

Collected data will be processed using advanced analytics and AI-driven models to:

» Identify trends and anomalies.

» Predict high-risk zones for mangrove degradation or loss.

» Simulate the impacts of natural disasters like tsunamis, sea-level rise, and cyclones on mangrove forests.

c) Risk Assessment Frameworks

WMC will adopt standardized risk assessment tools to evaluate:

» Vulnerability of different mangrove zones.

» Socioeconomic impacts of mangrove degradation on communities.

» Readiness and resilience capacity of countries and stakeholders.

d) Early Warning Alerts

A key function of WMC is to issue early warning alerts through:

» Dashboard notifications.

» SMS/email alerts for authorized stakeholders.

» Public information campaigns when necessary.

These alerts will enable rapid response measures, such as enforcement, restoration mobilization, or temporary closure of vulnerable areas.

3. International and Local Coordination

WMC's early warning system is designed to work in collaboration with local governments, community networks, and international partners, ensuring that:

- Local knowledge and ground-level observations are integrated.
- Alerts are context-sensitive and actionable.
- Response protocols are aligned across regions and institutions.

4. Capacity Building and Community Involvement

WMC will also play a role in building the capacity of partner countries and communities by:

- Offering training on risk identification and response.
- Distributing educational materials on threats to mangroves.
- Supporting community-based monitoring and disaster preparedness programs.

5. Continuous Improvement

The risk management system will be adaptive, constantly improving based on:

- Lessons learned from past events.
- User feedback and system evaluations.
- Technological advancements.

In 2024, the Government of Indonesia, as a host country and a key player in global mangrove conservation, presented a comprehensive vision for an integrated information system for the World Mangrove Center (WMC). This initiative reflects Indonesia's strong commitment to strengthening international collaboration, advancing sustainable environmental governance, and positioning the WMC as a global data hub for mangrove ecosystems.

This system enables the World Mangrove Center (WMC) to manage global mangrove data in an organized and efficient way. By centralizing information from various countries and internal studies, the WMC can maintain a single, unified database. Through standardization, all data is formatted consistently, making it easier to compare, analyze, and use across different regions. Finally, the system allows for effective dissemination of data through a dashboard, ensuring that researchers, policymakers, and conservationists can access up-to-date and reliable information. This supports better decision-making, more accurate research, and stronger conservation efforts for mangrove ecosystems worldwide.

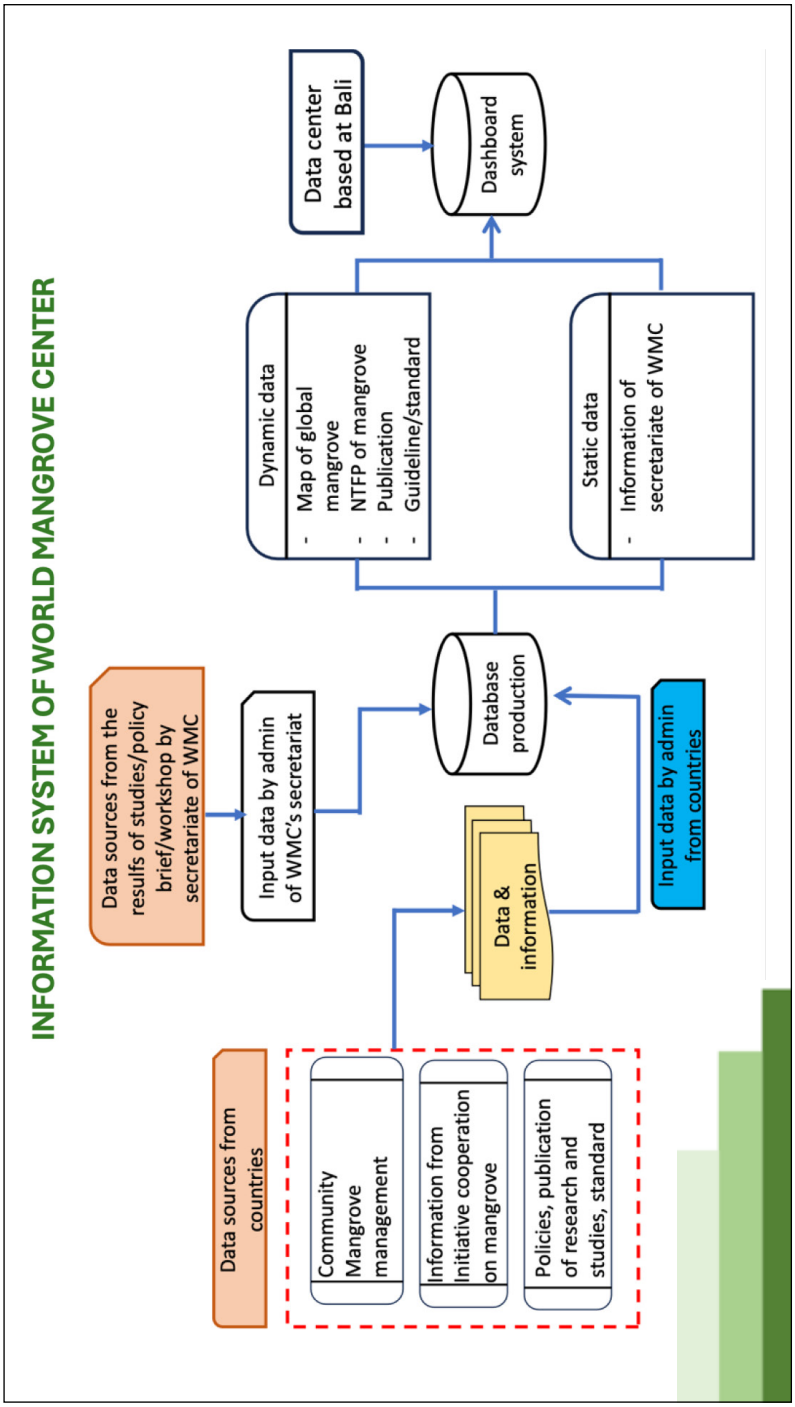

Figure 2. The image illustrates the Information System of the World Mangrove Center (WMC)

(KLHK RI, 2024)

Table 3. Components of Information System in WMC

Component	Explanation
Data Sources from Countries	Includes three subcategories: - Community Mangrove Management - Information from Initiative Cooperation on Mangrove - Policies, publication of research and studies, standard
Data Sources from WMC Secretariat	Derived from results of studies, policy briefs, and workshops conducted by the WMC Secretariat.
Input by Admins (Countries & Secretariat)	Data from countries and WMC Secretariat are input by designated administrators.
Data & Information	Initial repository where raw data from all sources is collected before processing.
Database Production	Central phase where input data is processed and stored in a structured database.
Dynamic Data	Includes data that is regularly updated: - Global mangrove map - Mangrove NTFP (Non-Timber Forest Products) - Publications - Guidelines and standards
Static Data	Includes permanent information such as details about the WMC Secretariat.
Dashboard System	Interface for accessing processed data. Presents both static and dynamic data in an organized format.
Data Center Based at Bali	Physical or cloud-based location where the dashboard system and central database are hosted.

V. THE CONNECTIVITY OF WMC WITH OTHER CENTERS IN INDONESIA

The World Mangrove Center (WMC) will be supported with a wide range of data and information, including inputs derived from best practices in mangrove management and from information centers established through international cooperation initiatives, particularly those involving Indonesia and partner countries.

Currently, Indonesia is actively engaged in two key international collaborations:

1. The Mangrove Information Center (MIC) – developed in collaboration with Japan.
2. The International Mangrove Research Center (IMRC) – developed in partnership with the United Arab Emirates (UAE).

These two centers in Bali (MIC and IMRC) are being established as centralized hubs or pools of data and information. Their purpose is to gather, store, and manage comprehensive information on mangrove ecosystems, including scientific research, field data, restoration techniques, community practices, and policy insights.

Both MIC and IMRC are designed to be digitally connected and interoperable with the WMC's information system. Once integrated, they will serve as official data and information

sources for WMC, enriching the Center's global database and strengthening its role as the leading platform for mangrove knowledge exchange, research collaboration, and policy support.

This integrated approach ensures that WMC benefits from the latest innovations, cross-country experiences, and scientific findings, thereby enhancing its capacity to support global mangrove conservation and sustainable management efforts.

The establishment of the World Mangrove Center (WMC) is guided by nine strategic priorities aimed at building a strong, sustainable, and globally connected institution for mangrove conservation. These priorities reflect both institutional development and collaborative engagement across countries and sectors.

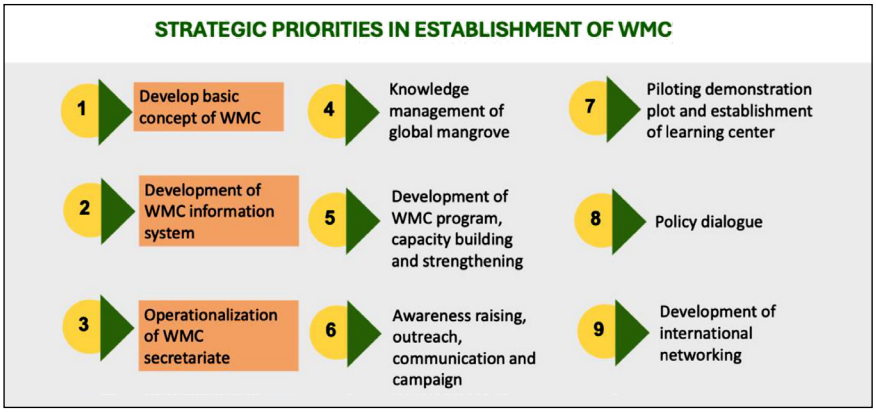

Figure 3. Strategic Priorities in Establishment of WMC

(KLHK RI, 2024)

1. Develop Basic Concept of WMC

   This initial priority focuses on formulating the vision, mission, objectives, and institutional framework of the WMC. It involves defining the Center's mandate, governance

structure, functional roles, and strategic direction. A strong foundational concept ensures that WMC can operate with clarity and purpose.

2. Development of WMC Information System

   An integrated data and information system is crucial for storing, managing, and sharing global mangrove data. This system will serve as a platform for collaboration, knowledge exchange, and evidence-based decision-making, supporting research, policy, and monitoring efforts worldwide.

3. Operationalization of WMC Secretariat

   This priority involves the establishment and staffing of the WMC Secretariat, which acts as the administrative and coordinating body. It includes defining roles, building institutional capacity, and ensuring smooth day-to-day operations to manage WMC activities effectively.

4. Knowledge Management of Global Mangrove

   To position WMC as a global knowledge hub, this strategy emphasizes the collection, analysis, and dissemination of mangrove-related knowledge, including best practices, research findings, traditional wisdom, and innovative technologies. It ensures that valuable insights are accessible and can be applied globally.

5. Development of WMC Program, Capacity Building, and Strengthening

   This includes designing and implementing structured programs for training, technical assistance, and institutional strengthening. The aim is to build the capacity of member countries, local communities, and stakeholders to effectively conserve and manage mangrove ecosystems.

6. Awareness Raising, Outreach, Communication, and Campaign

   A strong communication strategy is essential to build public support and stakeholder engagement. This priority focuses on educational campaigns, media engagement, publications, and outreach activities that raise awareness of mangrove conservation and the role of WMC.

7. Piloting Demonstration Plot and Establishment of Learning Center

   The WMC will set up demonstration sites and learning centers to showcase effective mangrove management and restoration techniques. These sites serve as hands on training grounds for researchers, practitioners, and policymakers.

8. Policy Dialogue

   WMC aims to facilitate regional and international policy dialogue on mangrove governance. This includes organizing forums, policy roundtables, and discussions to harmonize approaches, influence global environmental agendas, and support implementation of conventions like Ramsar and the Paris Agreement.

9. Development of International Networking

   Global collaboration is key to WMC's success. This priority focuses on establishing networks with countries, research institutions, international organizations, and NGOs to share data, promote innovation, and foster cooperative projects for mangrove conservation.

VI.

WORLD MANGROVE CENTER

The World Mangrove Center (WMC), initiated and hosted by Indonesia, represents a global initiative to promote the sustainable management, conservation, and restoration of mangrove ecosystems. Positioned as an international platform, the WMC is designed to become the central hub for mangrove knowledge, collaboration, innovation, and policy support.

Indonesia, home to the largest area of mangrove forests in the world, plays a crucial leadership role in global mangrove conservation efforts. The establishment of the WMC in Indonesia is not only a recognition of the country's ecological wealth but also a demonstration of its commitment to climate action, biodiversity protection, and sustainable coastal development.

A. World Mangrove Center Teluk Balikpapan – East Kalimantan

The launch of the World Mangrove Center in Teluk Balikpapan (WMCTB) took place in Rumpin, Bogor, as part of the main event for World Environment Day 2022, officiated by the President of the Republic of Indonesia, Joko Widodo. WMCTB is part of a global initiative supported by the Governments of Indonesia and Germany through the Ministry of Environment and Forestry (KLHK) and CIFOR-ICRAF. Its objective is to serve as an international

collaboration hub for sustainable mangrove management, to provide scientific data, and to support ecosystem-based climate action.

The development of WMCTB is one of the activities under the cooperation between the Indonesian and German Governments through the Forest Programme VI: Protection of Mangrove Forests, with the Environmental and Forestry Instrument Standardization Agency as the Executing Agency.

WMCTB, located in Graha Indah Subdistrict, North Balikpapan District, East Kalimantan, is a mangrove conservation area managed by the local community. The area functions as a center for education, conservation, and ecotourism, integrating environmental preservation with the economic empowerment of coastal communities.

Originally established in the early 2000s in response to mangrove degradation that triggered natural disasters such as tornadoes and extreme temperatures, the Graha Indah Mangrove Center has evolved into a model of community-based ecotourism. This initiative involves collaboration between local communities, local governments, and the private sector—including companies such as PT Pertamina and Total E&P—in efforts to restore and conserve mangrove ecosystems.

(a)

(b)

Figure 4. (a) WMC in Teluk Balikpapan; (b) President of the Republic of Indonesia, Joko Widodo, inaugurated the Rumpin Nursery in 2022

B. Mangrove Information Centre - Bali

Taman Hutan Raya or *Tahura* Ngurah Rai in Bali was officially designated as a conservation area in 1992 through the Decree of the Minister of Forestry No. 765/KPTS-II/1992, dated August 26, 1992. However, mangrove conservation efforts in the area began as early as the 1970s, when preservation initiatives for the coastal region of Benoa Bay were initiated in response to growing development pressures in Denpasar and Badung.

The World Mangrove Center (WMC) in Bali is an international initiative aimed at becoming a center of excellence for the conservation, research, and sustainable management of mangrove ecosystems. The center is located within Ngurah Rai Grand Forest Park (*Taman Hutan Raya* or Tahura) in Denpasar, Bali, which is the largest mangrove conservation area in Bali, covering approximately 1,158 hectares. Related Project in this WMC is Mohamed bin Zayed–Joko Widodo International Mangrove Research Center

As part of the WMC's development, Indonesia is collaborating with the United Arab Emirates (UAE) to establish the Mohamed bin Zayed–Joko Widodo International Mangrove Research Center within Tahura Ngurah Rai. The research center will occupy an area of 2.5 hectares.

The park spans parts of Denpasar City and Badung Regency, including villages such as Sanur Kauh, Sidakarya, Serangan, Kuta, and Tanjung Benoa. MIC is home to 33 species of mangroves and over 300 species of fauna, including various birds, reptiles, and aquatic organisms. The area serves critical ecological functions as a natural barrier against coastal erosion, a carbon sink, and a vital support system for the livelihoods of surrounding coastal communities.

Figure 5. The Groundbreaking of MBZ-JKW IMRC in Bali in 2024

C. World Mangrove Center – West Kalimantan

As a region located on the western part of Indonesia's second-largest island, West Kalimantan (*Kalimantan Barat* or *Kalbar*) has declared itself part of the "lungs of the Earth" since 2007 through the Heart of Borneo (HoB) initiative, in cooperation with Malaysia and Brunei Darussalam. It earned this title because its 5.4 million hectares of tropical rainforest produce about 40% of the Earth's oxygen.

The island is also home to 10 endemic primate species, over 350 bird species, 150 reptile and amphibian species, and 10,000 plant species. Between 2007 and 2010, 123 new species were discovered in this region, underlining its importance as a global biodiversity hotspot.

In addition, West Kalimantan has approximately 177,023.738 hectares of mangrove forests. These mangrove areas are spread across five regencies: Kubu Raya, Sambas, Mempawah, North Kayong, and Ketapang.

The mangrove biodiversity in West Kalimantan is both unique and rare. Notably, it is home to several rare tree species such as:

- *Bruguiera hainesii* (commonly known as *Bakau Mata Buaya*), one of the rarest mangrove species in the world. This species is critically endangered and found only in Singapore, Malaysia, Papua New Guinea, and Indonesia.
- *Kandelia candel* (*Lenggadai Betina*), which is endemic to Kalimantan and Sumatra.

West Kalimantan's mangrove ecosystems are also critical habitats for rare and endangered wildlife species, including:

- The Proboscis monkey (*Nasalis larvatus*), known locally as the *Bekantan* or "Dutch monkey", which is endemic to Borneo and classified as endangered.
- The Irrawaddy dolphin (*Orcaella brevirostris*), also known as *Pesut*, a rare freshwater dolphin species found in estuarine and coastal mangrove waters.

A formal proposal to establish a World Mangrove Center was submitted through the West Kalimantan Environment and Forestry Agency Letter No. 090/1072/DLHK-V/KSDAE, dated August 12, 2020, requesting support for the initiative. The center is planned to be located in Kubu Raya, which has the largest mangrove area in West Kalimantan. Out of the total mangrove area in West Kalimantan, Kubu Raya covers approximately 129,604.125 hectares, making it the most significant mangrove region in the province. This area also boasts high biodiversity, including:

- 33 true mangrove species out of the 40 species found in Indonesia, highlighting its global ecological importance.
- Rare and endemic flora such as *Bruguiera hainesii*.
- Fauna like the Bekantan and *Irrawaddy dolphin*, both of which rely on healthy mangrove habitats.

The establishment of the World Mangrove Center in Kubu Raya would not only strengthen conservation efforts but also support scientific research, community-based ecotourism, climate resilience, and sustainable coastal development.

Figure 6. Mangrove Condition in Batu Ampar Kubu Raya Regency

VII.

KEY CHALLENGES

The development of the World Mangrove Center (WMC) faces a number of complex and interrelated challenges that stem from its strategic role in environmental conservation, international climate commitments, and community-based sustainability. One of the foremost challenges lies in ensuring the long-term sustainability of the WMC's programs. Mangrove restoration and conservation efforts are inherently long-term in nature, often requiring 10 to 30 years to deliver measurable ecological benefits. This means that short-term funding cycles or project-based financing mechanisms are inadequate. Without a sustainable financial framework, there is a significant risk that the WMC could stall after its initial phases, such as pilot projects or infrastructure development, leaving long-term objectives unmet.

WMC also plays a key role in fulfilling Indonesia's global environmental commitments, including its Nationally Determined Contributions (NDCs) under the Paris Agreement, the Sustainable Development Goals (SDGs), and its participation in international blue carbon initiatives. Meeting these commitments requires not only technical capability but also access to global financial mechanisms, such as result-based financing and carbon markets. However, leveraging such mechanisms presents its own set of challenges, including the need for robust monitoring, reporting,

and verification systems that meet international standards, as well as building institutional capacity to engage with global financial actors.

In addition, one of the structural challenges is creating an enabling environment for multi-stakeholder collaboration. Effective mangrove management and the operation of the WMC require strong cooperation between central and local governments, international organizations, private sector actors, and local communities. Each of these stakeholders has different motivations, expectations, and governance frameworks, which must be harmonized through transparent and inclusive financing and operational models. The ability to build trust among these parties and coordinate their contributions is fundamental but often difficult in practice.

Finally, the risk of over-reliance on a single funding source, such as government budgets (APBN), poses a significant threat to the WMC's resilience. Fiscal uncertainties, political changes, or economic crises could jeopardize the continuity of funding. Therefore, developing a diversified funding portfolio—comprising government funds, philanthropic contributions, private sector investment, and carbon finance—is critical to mitigate financial risks and ensure institutional stability over the long term.

These challenges, while substantial, highlight the importance of thoughtful planning, innovative financing, and robust governance in transforming the WMC into a global center of excellence for mangrove conservation and sustainable development.

Figure 7. Key Challenges in Developing of WMC

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